Given: Due:	Further Maths (Core Pure) HW 20	Name: MARK SCHEME

Chapter 4.5: Roots of polynomials (inc. transformations of roots)

Q1. The cubic equation

$$2x^3 + 6x^2 - 3x + 12 = 0$$

has roots α, β and γ.

Without solving the equation, find the cubic equation whose roots are $(\alpha + 3)$, $(\beta + 3)$ and $(\gamma + 3)$, giving your answer in the form $pw^3 + qw^2 + rw + s = 0$, where p, q, r and s are integers to be found.

(5)

This question was successfully attempted and full marks were achieved by the majority of candidates. Although a topic new to the specification, this type of question has been seen numerous times in preparatory material and is a good indicator that candidates are already adapting well to the new content.
In contrast to the previous series, where the two method were both commonly used, responses to the question this year had a strong bias towards the main scheme, which is simpler to carry out and often more successful. The majority of candidates who selected this method scored 4 or 5 marks. In contrast candidates who used the sum/product of roots approach were much more prone to error and likely to score only 3.
In both of these methods, mistakes were seen in the manipulation involved but candidates demonstrated a sound knowledge of the process to be followed. Omission of the "=0" or use of alternative variables was rare.

Question	Scheme	Marks	AOs
	$\{w = x+3 \Rightarrow\}\ x = w-3$	B1	3.1a
	$2(w-3)^3 + 6(w-3)^2 - 3(w-3) + 12\ (=0)$	M1	1.1b
	$2w^3 - 18w^2 + 54w - 54 + 6(w^2 - 6w + 9) - 3w + 9 + 12 (=0)$		
	$2w^3 - 12w^2 + 15w + 21 = 0$	M1	3.1a
	(So $p = 2$, $q = -12$, $r = 15$ and $s = 21$)	A1 A1	1.1b 1.1b
		(5)	
ALT 1	$\alpha + \beta + \gamma = -\frac{6}{2} = -3,\ \alpha\beta + \beta\gamma + \alpha\gamma = -\frac{3}{2},\ \alpha\beta\gamma = -\frac{12}{2} = -6$	B1	3.1a
	sum roots $= \alpha + 3 + \beta + 3 + \gamma + 3$	M1	3.1a
	$= \alpha + \beta + \gamma + 9 = -3 + 9 = 6$		
	pair sum $= (\alpha+3)(\beta+3) + (\alpha+3)(\gamma+3) + (\beta+3)(\gamma+3)$		
	$= \alpha\beta + \alpha\gamma + \beta\gamma + 6(\alpha+\beta+\gamma) + 27$		
	$= -\frac{3}{2} + 6\times -3 + 27 = \frac{15}{2}$		
	product $= (\alpha+3)(\beta+3)(\gamma+3)$		
	$= \alpha\beta\gamma + 3(\alpha\beta + \alpha\gamma + \beta\gamma) + 9(\alpha+\beta+\gamma) + 27$		
	$= -6 + 3\times -\frac{3}{2} + 9\times -3 + 27 = -\frac{21}{2}$		
	$w^3 - 6w^2 + \frac{15}{2}w - \left(-\frac{21}{2}\right)\ (=0)$	M1	1.1b
	$2w^3 - 12w^2 + 15w + 21 = 0$ (So $p = 2$, $q = -12$, $r = 15$ and $s = 21$)	A1 A1	1.1b 1.1b
		(5)	
		(5 marks)	

Notes		
	B1	Selects the method of making a connection between x and w by writing $x = w - 3$
	M1	Applies the process of substituting their $x = aw \pm b$ into $2x^3 + 6x^2 - 3x + 12\ (=0)$ So accept e.g. if $x = \frac{w}{3}$ is used.
	M1	Depends on having attempted substituting either $x = w - 3$ or $x = w + 3$ into the equation. This mark is for manipulating their resulting equation into the form $pw^3 + qw^2 + rw + s (=0)$ ($p \neq 0$). The "= 0" may be implied for this.
See note	A1	At least three of p, q, r and s are correct in an equation with integer coefficients. (need not have "= 0")
	A1	Correct final equation, including "=0". Accept integer multiples.

ALT 1	B1	Selects the method of giving three correct equations each containing α, β and γ.
	M1	Applies the process of finding sum roots, pair sum and product.
	M1	Applies w^3 − (their sum roots)w^2 + (their pair sum)w − (their product) (= 0) Must be correct identities, but if quoted allow slips in substitution, but the "=0" may be implied.
See note	A1	At least three of p, q, r and s are correct in an equation with integer coefficients. (need not have "=0")
	A1	Correct final equation, including "=0". Accept multiples with integer coefficients.

Note: may use another variable than w for the first four marks, but the final equation must be in terms of w
Notes: Do not isw the final two A marks – if subsequent division by 2 occurs then mark the final answer.

Q2.

The three solutions of the cubic equation

$$x^3-2x^2+3x+1=0 \quad x\in\mathbb{R},$$

are denoted by α, β and γ.

Find a cubic equation with integer coefficients whose solutions are

$$2\alpha-1,\ 2\beta-1 \text{ and } 2\gamma-1.$$

(4)

$x^3-2x^2+3x+1=0$

LET THE SOLUTION OF THE NEW CUBIC BY X

$X=2x-1 \iff x=\frac{X+1}{2}$

$\left(\frac{X+1}{2}\right)^3-2\left(\frac{X+1}{2}\right)^2+3\left(\frac{X+1}{2}\right)+1=0$

$\frac{1}{8}(X^3+3X^2+3X+1)-\frac{1}{2}(X^2+2X+1)+\frac{3}{2}(X+1)+1=0$

$(X^3+3X^2+3X+1)-4(X^2+2X+1)+12(X+1)+8=0$

$X^3+3X^2+3X+1-4X^2-8X-4+12X+12+8=0$

$\therefore X^3-X^2+7X+17=0$

Q3.

The roots of the cubic equation

$$16x^3 - 8x^2 + 4x - 1 = 0 \quad x \in \mathbb{R},$$

are denoted in the usual notation by α, β and γ.

Find a cubic equation, with integer coefficients, whose roots are

$$\tfrac{4}{3}(\alpha - 1), \ \tfrac{4}{3}(\beta - 1) \ \text{and} \ \tfrac{4}{3}(\gamma - 1).$$

(6)

- USING A SUBSTITUTION HERE

$$y = \tfrac{4}{3}(x-1)$$
$$3y = 4x - 4$$
$$4x = 3y + 4$$

- MANIPULATE THE CUBIC FOR SIMPLICITY

$$\Rightarrow 16x^3 - 8x^2 + 4x - 1 = 0$$
$$\Rightarrow 64x^3 - 32x^2 + 16x - 4 = 0 \quad \times 4$$
$$\Rightarrow (4x)^3 - 2(4x)^2 + 4(4x) - 4 = 0$$
$$\Rightarrow (3y+4)^3 - 2(3y+4)^2 + 4(3y+4) - 4 = 0$$

Now:

$$\boxed{\begin{aligned}(A+B)^3 &= A^3 + 3A^2B + 3AB^2 + B^3\\ (3y+4)^3 &= 27y^3 + 3(3y)^2\times 4 + 3(3y)\times 16 + 64\\ &= 27y^3 + 108y^2 + 144y + 64\end{aligned}}$$

$$\Rightarrow 27y^3 + 108y^2 + 144y + 64 - 2(9y^2 + 24y + 16) + 12y + 16 - 4 = 0$$

$$\Rightarrow \left.\begin{array}{r}27y^3 + 108y^2 + 144y + 64\\ -18y^2 - 48y - 32\\ 12y + 12\end{array}\right\} = 0$$

$$\Rightarrow 27y^3 + 90y^2 + 108y + 44 = 0$$

Q4.

The polynomial $x^4 + x^3 - x + 12 = 0$ has roots $\alpha, \beta, \gamma, \delta$. Find the polynomial with roots $\alpha^2, \beta^2, \gamma^2, \delta^2$. (4)

Answer

$y = x^2$	State the substitution.
$x^4 + 12 = x - x^3$	Rearrange so that both sides when squared give even terms.
$x^8 + 24x^4 + 144 = x^2 - 2x^4 + x^6$	Square both sides.
$x^8 - x^6 + 26x^4 - x^2 + 144 = 0$	Simplify.
$y^4 - y^3 + 26y^2 - y + 144 = 0$	Use $x^2 = y$.

Q5.

The equation $x^4 - 6x^3 - 73x^2 + kx + m = 0$ has two positive roots, α, β and two negative roots γ, δ. It is given that $\alpha\beta = \gamma\delta = 4$.

(i) Find the values of the constants k and m. (4)

(ii) Show that $(\alpha + \beta)(\gamma + \delta) = -81$. (3)

(iii) Find the quadratic equation which has roots $\alpha + \beta$ and $\gamma + \delta$. (2)

(iv) Find $\alpha + \beta$ and $\gamma + \delta$. (2)

(v) Show that $\alpha^2 - 3(1+\sqrt{10})\alpha + 4 = 0$, and find similar quadratic equations satisfied by β, γ and δ. (3)

(i) $x^4 - 6x^3 - 73x^2 + kx + m = 0$

$\alpha\beta\gamma\delta = m$

$4 \times 4 = m$

$m = 16$

$\alpha\beta\gamma + \beta\gamma\delta + \gamma\delta\alpha + \delta\alpha\beta = -k$

$4\gamma + 4\beta + 4\alpha + 4\delta = -k$

$k = -4(\alpha + \beta + \gamma + \delta)$

$k = -4 \times -\dfrac{b}{a}$

$k = 4 \times -6 = -24$

[4]

(ii) $(\alpha + \beta)(\gamma + \delta) = \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta$

$= \sum \alpha\beta - (\alpha\beta + \gamma\delta)$

$= \dfrac{c}{a} - (4 + 4)$

$= -73 - 8$

$= -81$

[3]

(iii) Sum of roots of quadratic equation $= \alpha + \beta + \gamma + \delta = 6$

Product of roots of quadratic equation $= (\alpha + \beta)(\gamma + \delta) = -81$

Quadratic equation is therefore $x^2 - 6x - 81 = 0$

[2]

(iv) $x^2 - 6x - 81 = 0$

$x = \dfrac{6 \pm \sqrt{36 - 4 \times 1 \times -81}}{2} = 3 \pm 3\sqrt{10}$

$\alpha + \beta = 3 + 3\sqrt{10}$

$\gamma + \delta = 3 - 3\sqrt{10}$

[3]

(v) $\alpha + \beta = 3 + 3\sqrt{10}$ and $\alpha\beta = 4$

so α and β are the roots of the quadratic equation

$x^2 - 3(1 + \sqrt{10})x + 4 = 0$

so α satisfies $\alpha^2 - 3(1 + \sqrt{10})\alpha + 4 = 0$

and β satisfies $\beta^2 - 3(1 + \sqrt{10})\beta + 4 = 0$

$\gamma + \delta = 3 - 3\sqrt{10}$ and $\gamma\delta = 4$

so γ and δ are the roots of the quadratic equation

$x^2 - 3(1 - \sqrt{10})x + 4 = 0$

so γ satisfies $\gamma^2 - 3(1 - \sqrt{10})\gamma + 4 = 0$

and δ satisfies $\delta^2 - 3(1 - \sqrt{10})\delta + 4 = 0$